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Optically reconfigurable physical unclonable functions based on 2D MoS_2 ring-oscillator arrays for attack-resistant hardware authentication

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Abstract: Hardware-level security is crucial for establishing trust in the rapidly expanding Internet-of-Things (IoT) and edge computing systems. A promising approach employs physical unclonable functions (PUFs) which leverage intrinsic process variations to generate unique and irreproducible identifiers. However, conventional silicon-based PUFs often suffer from limited entropy and lack of reconfigurability, becoming vulnerable to machine learning attacks. Here, we present an optically reconfigurable PUF based on a 64-cell array of five-stage ring oscillators fabricated from wafer-scale monolayer MoS_2. The system exhibits spectrally selective frequency shifts under red, green, and blue (RGB) illumination, establishing a dynamic optical entropy dimension that enables on-demand, reversible rekeying without hardware modification. We develop a robust key-generation pipeline combining within-chip normalization, random-projection q-ary quantization, and the Secure Hash Algorithm 256 (SHA-256) privacy amplification, followed by hash-based message authentication code (HMAC)-based key derivation. The resulting keys demonstrate near-ideal uniformity (~50%) and inter-device Hamming distance (~0.677 at q = 3, 39 ~0.761 at q = 4), while remaining resilient to advanced machine learning attacks ($\leqslant$ 52% accuracy). We further demonstrate image encryption and authentication with noise-like ciphertexts and reliable tamper detection. This work introduces a promising class of material-intrinsic, optically addressable security primitives for trusted edge computing applications.

KEYWORDS: MoS_2 ring oscillators, optoelectronic reconfigurability, cryptography, physical unclonable function, attack resistance

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Introduction

As digital systems play crucial roles in our daily life[1], ensuring hardware-level security is crucial for building trustworthy cyber-physical environments[2] especially in Internet of Things (IoT) and edge computing deployments, where devices operate autonomously in potentially untrusted or hostile conditions[3]. Traditional software-based cryptographic protections are insufficient alone, particularly against physical attack vectors or invasive access[4]. Physical Unclonable Functions (PUFs) have emerged as an efficient hardware security primitive that utilizes uncontrollable process-induced variations to generate device-specific digital fingerprints[5-7]. Among various architectures, ring oscillator (RO)-based PUFs offer advantages in power efficiency, compactness, and real-time responsiveness[8]. However, conventional RO-PUFs face notable limitations, including low entropy density, static challenge-response spaces, and increasing susceptibility to machine learning attacks, which collectively restrict their robustness and applicability in scalable, real-world security systems[6, 9-11].

Recent advances in two-dimensional (2D) semiconductors, particularly monolayer MoS_2, offer promising approaches for hardware-level entropy design[12, 13]. These emergent semiconductors exhibit excellent electrical properties, optical sensitivity, and compatibility with wafer-scale fabrications, thereby enabling the development of reconfigurable, optoelectronic PUF architectures[14-17]. Previous implementations of 2D-material-based PUFs have largely relied on static entropy sources, such as growth-induced morphological disorder, hetero-nanostructure randomness or transistor-level threshold voltage variability, to generate unclonable responses[17-20]. These approaches successfully harness material- and process-induced disorder to broaden the accessible parameter space. However, the derived challenge-response pairs (CRPs) from device-level variability remain essentially fixed after fabrication, offering limited post-deployment reconfigurability and a static challenge space that can become susceptible to modeling over time. Notably, some recent 2D-material security primitives have demonstrated intrinsic resilience to machine-learning attacks enabled by physics-level stochasticity or device reconfigurability[21-23], even without explicit post-processing steps [12, 13]. Nevertheless, a significant need remains for a

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PUF system that integrates material-level tunability, multi-modal entropy, and cryptographic robustness into a unified, scalable platform[24-26].

In contrast to static entropy sources, dynamic and reconfigurable entropy dimensions such as wavelength- and intensity-dependent photo-modulation can substantially expand the challenge space and enable renewable cryptographic keys without altering the physical hardware. Here, we introduce an optically reconfigurable ring-oscillator PUF, in which illumination wavelength and intensity act as dynamic challenge inputs, reshaping the RO frequency landscape to enable reversible rekeying and multidimensional CRP mappings. This work demonstrates an optically reconfigurable, cryptographically enhanced RO-PUF architecture based on wafer-scale monolayer MoS_2 (Figure 1). The system integrates a 64-cell array of five-stage ring oscillators whose frequencies exhibit wavelength-selective modulation under red, green, and blue (RGB) light, enabling a dual-entropy source through both intrinsic electrical variability and extrinsic optical perturbation. In our PUF, CRP describes the mapping between an input challenge C and its corresponding output response R(C). In our optically reconfigurable RO-PUF, the optical stimulus modulates the oscillation frequencies of the MoS_2 ring oscillator array, producing a frequency vector that is subsequently processed by a fixed response-generation pipeline. This processing converts the frequency features into a binary codeword r(C), which we define as the PUF response. The CRP is therefore represented as the pair (C, r(C)), while an optional cryptographic hash (SHA-256) can further derive a final key K(C) for secure sensing and encryption applications. The nominal capacity of this system, assuming all bits in the pre-hash codeword are independent, is 2^{256}, representing the theoretical key space. Accounting for statistical correlations among bits and bit reliability across repeated measurements, the actual, experimentally validated capacity is approximately 2^{235}, reflecting the true degrees of freedom available for secure key generation (Supplementary Table 2&3). This distinction highlights both the large addressable CRP space and the practical entropy achievable in our optical RO-PUF[27]. We treat (C, r(C)) as the CRP. SHA-256 can then be optionally applied as a privacy-amplification step to obtain the final key K(C). This processing pipeline integrates Secure Hash Algorithm 256 (SHA-256) hashing and hash-based

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message authentication code (HMAC)-based key derivation yielding reproducible 32-bit keys with near-ideal uniformity (~50%) and high inter-device uniqueness (Hamming distance (HD) ≈ 0.677 at q = 3, 0.761 at q = 4). Machine learning attack evaluations on up to 10^5 CRPs confirm strong modeling resilience, with advanced classifiers (e.g., CNNs, gradient boosting) achieving prediction accuracies close to random guessing (~50%)[22]. These properties support in-sensor encryption, session-level key diversification, and device authentication with minimal overhead. Compared to previous silicon or 2D-material PUFs[5, 23, 28, 29], our platform uniquely combines optical-electrical reconfigurability, statistical optimality, and cryptographic robustness, establishing a scalable, material-intrinsic security primitive for intelligent edge computing and next-generation trusted hardware.

Results

Fabrication of RO-Array Design

We utilize the superior electrical properties and optical responsiveness of monolayer MoS_2[16] to construct scalable RO arrays with high stability and reproducibility-key prerequisites for generating stable and secure cryptographic keys. Supplementary Figure 1 presents the synthesis and characterization of wafer-scale monolayer MoS_2 films, which serve as the active channel material for the RO array. Continuous 2-inch monolayer MoS_2 films are epitaxially grown on c-plane sapphire substrates via an oxygen-assisted chemical vapor deposition (O-CVD) process[16], exhibiting uniform optical contrast across the entire wafer (Supplementary Figure 1a). Atomic force microscopy (AFM) measurements confirm an atomically flat surface with a root-mean-square roughness below 0.4 nm[15] (Supplementary Figure 1b), indicating high surface uniformity. Raman spectra display sharp E_{2g} (384 cm^{-1}) and A_{1g} (403 cm^{-1}) modes with a peak separation of 19 cm^{-1} (Supplementary Figure 1c) [15], confirming the monolayer nature. Photoluminescence (PL) spectra exhibit a strong A-exciton peak centered at 1.88 eV[15] (Supplementary Figure 1d), consistent with a direct bandgap and high crystalline quality. These results demonstrate that the

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O-CVD films combine wafer-scale uniformity with the optical and electronic properties required for large-scale device integration[30].

To ensure a minimal device-to-device variation in the RO array, we evaluated the electrical characteristics of 100 randomly selected MoS_2 field-effect transistors (FETs)[30] (Figure 2 & Supplementary Table 1) are evaluated. Transfer curves from representative devices (Figure 2a) reveal steep subthreshold slopes, on/off current ratios exceeding 10^8, and negligible hysteresis. Statistical analysis demonstrates tight parameter distributions: the field-effect mobility averages at 41.69 $cm^2\ V^{-1}\ s^{-1}$ with a maximum value of 51.69 $cm^2\ V^{-1}\ s^{-1}$ (Supplementary Figure 2a); the on/off ratio centers around 2.7×10^8 (Supplementary Figure 2b); the threshold voltage is narrowly distributed around −1.73 V (Figure 2b); and the current density clusters near 50 $\mu A\ \mu m^{-1}$ (Supplementary Figure 2c); and the average subthreshold swing is 208 $mV\ dec^{-1}$ (Supplementary Figure 2d). We also report mobility values after simple series-resistance de-embedding using the measured Rc 1.16 kΩ, $R_{\square}$ 1.04kΩ. These results indicate that the frequency variations in the RO array arise from intrinsic microscopic fluctuations rather than fabrication-induced defects[31], which is essential for generating reliable PUF entropy.

Based on the uniform device performance, we fabricate a 64-cell RO array, with each cell consisting of five cascaded inverters formed by paired MoS_2 FETs (Figure 2c). The die photograph shows the compact array layout, while the zoomed-in image reveals precise pattern definition and consistent device geometry across the array. The five-stage RO design provides a balance between oscillation frequency, signal stability, and fabrication complexity[31, 32], producing clean periodic waveforms suitable for high-quality entropy harvesting (Supplementary Note 1). The successful integration of monolayer MoS_2 devices into a dense RO array establishes a robust hardware platform for the optically reconfigurable and hash-enhanced PUF system described in the following sections.

RO Performance Characterizations

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The MoS_2-based ROs exhibits stable self-sustained oscillations in the absence of optical illumination[32] (Figure 2c). The normalized output voltage remained consistently bounded between 0 and 0.9 over a 0-20 ms observation window, with no observable baseline drift. This behavior indicates stable propagation delays (τ_p) across the inverter stages, reflecting reliable signal transmission dynamics. Such stability arises from robust carrier transport in the MoS_2 channels and negligible variation in parasitic capacitance or contact resistance under continuous operation[14]. Under increased gate bias (Figure 2d), the oscillation period decreases markedly, accompanied by steeper edge transitions and a duty cycle approaching 50 %[33]. These changes signify an enhanced channel conductivity and a reduced effective RC time constant, which collectively accelerating signal propagation through the inverter chain. This behavior is attributed to electrostatic modulation of the carrier density in the MoS_2 channel, which lowers the threshold voltage and increases the transconductance (g_m), thereby improving the switching speed of each inverter stage.

The frequency-power characteristics under different illumination wavelengths are shown in Figure 2e-f, revealing pronounced spectral selectivity. Under 650-nm laser excitation, the oscillation frequency increases from ~70.0 kHz at 0.3 normalized power (0.683 mW/mm^2) to ~550.0 kHz at full power (1.055 mW/mm^2), exhibiting the steepest tuning slope. For 530-nm laser excitation, the frequency rises from ~50.0 kHz to ~320.0 kHz, whereas 460-nm laser excitation yields only a modest increase from ~5.0 kHz to ~35.0 kHz. This wavelength dependence correlates with the optical absorption spectrum of MoS_2 and aligns with the PL spectra in Supplementary Figure 1d[16], confirming that 650-nm laser excitation directly couples to the A-exciton resonance, thereby accounting for the observed maximum tuning slope. This resonance enables efficient carrier generation with deeper penetration into the channel, facilitating more uniform conductivity modulation. In contrast, high-energy blue photons (460 nm) are strongly absorbed near the surface and contact regions, where rapid nonradiative recombination and localized heating reduce the net photoconductive gain[34]. Optical micrographs of the device under RGB illumination (Supplementary Figure 3) further confirm operational stability across the visible range, albeit with wavelength-dependent tuning sensitivity.

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Long-term stability tests (Supplementary Figure 4) demonstrate excellent operational reliability of the oscillator under dynamic optical modulation[35]. As shown in Figure. 2g, the oscillation frequency of the RO remains within 1.04-1.36 kHz over 72 h under fixed bias and ambient conditions, corresponding to a peak-to-peak drift of approximately 0.32 kHz. This result confirms stable multi-day operation of the MoS_2 RO. Specifically, we evaluated the output characteristics and RO performance across an extended temperature range of −20°C to +50°C (in ~20°C step increments) and under varying relative humidity levels from 0% to 100%, as summarized in Figure. 2h and Figure. 2i. The device maintains robust performance over the entire temperature range, exhibiting minor frequency shifts due to thermal modulation of carrier mobility, while key uniqueness and reproducibility remain largely unaffected. At higher humidity levels (≥60%), MoS_2 mobility slightly decreases, potentially causing minor fluctuations in response times and uniformity, nevertheless, the PUF remains fully functional under all tested conditions. In the dark state (Supplementary Figure 4a), the output signal remains stable and low-noise over a ± 500 s window. Under repeated illumination-removal cycles (Supplementary Figure 4b), the output voltage and oscillation frequency exhibit fully reversible transitions without notable degradation. The temporal response comprises a fast component, associated with instantaneous photocarrier generation and channel conductance modulation, and a slower component, likely arising from deep trap charging/discharging or dielectric polarization relaxation. The coexistence of these two timescales does not compromise frequency stability within a single oscillation period but governs the longer-term drift behavior under steady illumination. Collectively, these results collectively underscore the strong environmental robustness and operational stability of the MoS_2-based ring oscillator, affirming its suitability for reliable long-term deployment in hardware security applications. The demonstrated stability, spectral selectivity, and broadband tunability highlight the potential of MoS_2-based ROs as optically reconfigurable frequency sources for on-chip clock distribution, wavelength-multiplexed signal processing, and integrated optoelectronic sensing platforms[36].

Harvesting Physical Unclonable Function Keys

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The inherent device-to-device variations in oscillation frequency of the 8×8 array of five-stage MoS_2 ROs provide substantial entropy for hardware security applications[13]. Figure 2g presents the spatial frequency distribution maps of the array under four illumination conditions: dark (no illumination), red (650 nm), green (530 nm), and blue (460 nm). Under dark conditions, the measured frequencies range from approximately 0.1 to 0.9 MHz, exhibiting moderate spatial non-uniformity. Under illumination, the overall frequency increases and the spatial pattern changes distinctly in a wavelength-dependent manner[37]. Red light illumination produces the largest enhancement, with many cells exceeding 1.5 MHz, while green light yields intermediate enhancement, and blue light results in the smallest shift, consistent with the wavelength selectivity observed in single-device measurements. Some rows, such as those near index 6, exhibit consistently higher frequencies across all conditions, suggesting a minor systematic spatial bias likely originating from interconnect parasitics or local variations in contact resistance. These combined effects of global optical tuning and intrinsic microscopic variability form the statistical foundation for robust PUF responses[38].

To convert the frequency data into binary keys, we first apply a global thresholding method to classify each oscillator's output as '0' or '1' (Supplementary Figure 5a, Supplementary Note 2) yielding condition-specific PUF patterns for the dark, red, green, and blue states to directly capture the interplay between the intrinsic variability and optical modulation. However, marginal bits near the threshold are sensitive to small drifts, leading to reduced stability. To address this, we employ a quantile-based selection strategy that uses the first (Q25) and third (Q75) quartiles of the frequency distribution as adaptive thresholds (Supplementary Figure 5b). Bits within the Q25 to Q75 range are discarded to suppress instability, thereby improving uniformity and reliability while maintaining a high bit yield[39]. To assess the reproducibility of the proposed PUF, we calculate the normalized intra-HD from repeated measurements of the same CRP at the pre-hash response stage[27]. For each chip, 30 challenges are tested, and each CRP was read out 30 times under identical operating conditions. The average intra-HD of the raw response was 0.0605, which decreased to 0.0280 after quantile filtering (Q25/Q75) (Supplementary Table 4). This reduction indicates that

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quantile filtering improves repeat-read stability by excluding unstable comparisons near the decision boundary.

Entropy is further enhanced through two complementary strategies. First, responses from the four illumination conditions are combined via bitwise XOR to generate XOR-derived keys (Supplementary Figure 5c). This approach amplifies the effect of illumination-induced variability, pushing the inter-HD closer to the ideal 50 % while preserving low intra-HD[39]. Second, native PUF responses are processed using the SHA-256 cryptographic hash function to generate fixed-length 256-bit digests. The hashing process eliminates residual statistical bias, ensures cryptographic unpredictability, and permits key truncation without compromising randomness[40]. Both approaches independently enhance the security properties of the PUF keys, with XOR maximizes uniqueness through physical diversity, while SHA-256 ensures resistance to correlation-based inference.

Statistical evaluation of these methods across eight representative PUF configurations is shown in Supplementary Figure 6 comparing uniformity, uniqueness, and randomness metrics for various bit lengths and processing strategies (Supplementary Note 3). Uniformity remains close to the ideal 0.50 for most methods, with the Combined-Quantile 8-bit case showing a deviation (0.75) due to selective bit retention. Uniqueness is highest for the Threshold Method 4-bit and Combined-Median 4-bit configurations (0.51), while randomness reaches the maximum value of 1.00 for several cases, including the XOR-enhanced and SHA-256-enhanced schemes. These results quantitatively demonstrate that combining optical diversity with post-processing not only preserves statistical balance and inter-device independence but also maximizes the unpredictability of the generated keys[41].

To reliably extract cryptographic keys from the MoS_2 RO array, we implemented a multi-stage entropy harvesting and post-processing pipeline. Each device key is derived from a four-channel RO frequency vector, normalized within-chip using z-score standardization to minimize

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global bias and enhance cross-device comparability. The normalized vectors are then projected onto M = 512 independent Gaussian directions, yielding scalar responses that capture complementary aspects of intrinsic and optically induced variability. For each projection, data-driven equal-frequency thresholds (tertiles for q = 3 and quartiles for q = 4) are determined from a calibration cohort and used to quantize the responses into q-ary symbols. Candidate positions are subsequently ranked by deviation from uniformity and cross-correlation, with the K = 32 most balanced, low-correlated positions concatenated to form a 32-symbol q-ary codeword (A schematic illustration of this key generation pipeline is provided in Supplementary Figure 7). This design ensures that the selected symbols provide high entropy density and reproducibility while minimizing statistical bias. While a 32-bit pre-hash response length is used in this proof-of-concept demonstration, the proposed RO-PUF architecture is not inherently limited to 32 bits. For an array of N_{RO} oscillators, the number of available RO comparison pairs scales as $N_{pair} \propto N^2_{RO}$ (e.g., 2016 pairs for N_{RO}=64). Longer key material can be generated in several ways: by selecting a larger number of RO-pair comparisons (or quantized symbols) per challenge; by concatenating multiple response blocks across independent selection words and optical states (λ, P_{opt}); or by expanding the physical array itself. Since N_{pair} grows quadratically with array size, scaling the array directly increases the available entropy sources. Moreover, through stable-bit selection and optional error-tolerant regeneration, longer keys can maintain strong uniqueness and uniformity while preserving reproducibility.

Representative symbol maps (Figure 3a-d) illustrate that the extracted ternary and quaternary codewords distribute uniformly across both device index and symbol index, confirming that the quantization strategy suppresses spatial bias and yields balanced symbol occupancies. Inter-device independence was quantified through normalized inter-HD analysis (Figure 3e). The raw multi-level codewords yield mean inter-HD values of approximately two-thirds for ternary encoding and three-quarters for quaternary encoding, in agreement with the theoretical expectations for uniformly random M-ary alphabets. The histograms exhibit narrow spreads, indicating strong device separability and reliable uniqueness. To further suppress residual correlations and enable

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cryptographic applications, the serialized codeword is hashed using SHA-256, with the first 32 bits of the digest retained as the device key, while domain separation labels enable multiple independent keys per device. As shown in Figure 3f, hashing consolidates entropy into binary outputs with uniformity close to the ideal 0.50, uniqueness converging toward the theoretical optimum of 0.5 inter-HD, and randomness values approaching unity. The un-hashed PUF responses and the SHA-256-derived 256-bit keys are separately evaluated to distinguish the intrinsic randomness of the physical PUF from the statistical properties of the final extracted keys. The un-hashed responses are analyzed using response balance/uniformity, entropy-related metrics, and serial-dependence statistics to characterize the physical entropy source itself. The SHA-256-derived keys are then evaluated using the applicable NIST SP800-22 tests (Supplementary Table 5&6) to verify the statistical quality of the final cryptographic outputs. Accordingly, the raw-response statistics serve as evidence of intrinsic PUF randomness, whereas the SP800-22 results validate only the randomness of the final post-processed keys[27].

In the present work, the final SHA-256 key is not derived from a single raw noisy readout. Instead, under a given challenge and operating condition, the RO-PUF response is measured repeatedly, and the pre-hash response is stabilized by majority-vote processing, together with bit screening where applicable (Supplementary Figure 8). During enrollment, this procedure generated the enrolled pre-hash codeword (r_star). During regeneration, the same challenge and operating condition are applied, and the same repeated-read stabilization procedure is used to obtain the regenerated pre-hash codeword (r_hat). SHA-256 is then applied only after this stabilization step. Therefore, the final key is reproduced only when the stabilized pre-hash codeword is regenerated identically (r_hat = r_star). In other words, SHA-256 serves as a deterministic cryptographic mapping and does not itself improve response reliability.

The stabilization pipeline described above relies on repeated readouts and majority voting to suppress transient noise. To further assess whether the PUF can tolerate longer-term environmental variations that may shift the absolute oscillation frequencies, we characterize the RO stability under controlled humidity and temperature conditions. As shown in Figure 2i, the oscillation

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frequency of the unencapsulated RO decreases markedly with increasing relative humidity, whereas the encapsulated RO remains nearly constant over the same range, indicating that encapsulation can effectively suppresses humidity-induced drift[42]. Under temperature-controlled measurements (Figure 2h), the RO frequency remains stable over time at each fixed temperature, while its absolute value shifts systematically with temperature[43].For PUF operation, the response bit is determined by the relative ordering of a compared RO pair rather than the absolute frequency of an individual oscillator the response bit is determined by the relative ordering of a compared RO pair rather than by the absolute frequency of an individual oscillator. The observed systematic shift in absolute frequency with temperature (Figure 2h) does not necessarily alter the relative order of two oscillators unless their differential drift becomes comparable to the initial frequency margin. These device-level measurements therefore provide a preliminary assessment of environmental stability, indicating that the dominant temperature effect is a deterministic offset rather than random short-term fluctuations (Supplementary Note 4). Nevertheless, the present device-level stability, together with the stabilization pipeline described above, provides a solid foundation for constructing cryptographically robust keys from the MoS_2 RO arrays .

Building on this foundation, the combination of multi-ary quantization and cryptographic hashing provides a robust privacy amplification mechanism: multi-level extraction maximizes inter-device distances, while hashing transforms them into binary sequences with near-ideal statistical characteristics. For session-level cryptographic applications, such as encryption or message authentication codes (MACs), the hashed codewords are further expanded via a HMAC-based key derivation function (HKDF) seeded with per-session nonces, ensuring independence across sessions and application domains. The combination of intrinsic fabrication induced variability, spectrally controlled optical perturbations, and dual enhancement strategies enables the harvesting of high quality, reconfigurable PUF keys from the monolayer MoS_2 RO arrays. This platform offers a tunable, physical unclonable, and cryptographically robust hardware primitive, suitable for dynamic authentication protocols and next generation secure systems[13]. Using the implemented readout pipeline, we estimate an average generation time of∼5 ms for a 256-bit

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response, corresponding to a bit generation rate of ~51.2 kbps; the average acquisition power is ~2.93 nW at V_{DD} = 4V, yielding an energy cost of~57.2 fJ/bit (Supplementary Note 5).

Demonstration of PUF Secret Key Encryption and Decryption

To validate the practical applicability of the MoS_2 RO-PUF for secure key generation, we implemented proof-of-concept encryption and decryption experiments using PUF-derived keystreams (Figure 4). A compact 32-bit digest was extracted from each device (q-ary random-projection quantization followed by SHA-256 privacy amplification) and utilized as the seed for session keys. A fresh, per-image nonce is fed to HKDF to derive independent encryption and authentication keys; the former drives a SHA-256-CTR keystream and the latter an HMAC-SHA256 tag (encrypt-then-MAC), respectively. This arrangement preserves the hardware origin of entropy while enforcing modern, replay-safe, and integrity-checked operation at negligible computational cost. In a binary stream-cipher example of graphic encryption test (Figure 4a), a grayscale baboon image is thresholded to 1-bit symbols and XORed with the PUF-driven keystream, yielding a ciphertext that visually collapses to white noise and is quantified by a Geary's c = 0.99, i.e., negligible spatial correlation. Decryption via a second XOR recovers the original image exactly, evidencing both the bit-level reproducibility of the harvested key and the absence of systematic artifacts in the cryptographic front end. Figure 4b extends the procedure to quaternary symbols (2-bit grayscale quantization). The ciphertext again becomes noise-like with Geary's c = 1.00, and the perfect reconstruction verifies that domain-separated key derivation maintains independence across binary and multilevel streams while preserving whitening strength. The resulting ciphertext exhibits noise-like statistics with no discernible structural information, confirming the high entropy and uniformity of the native PUF key[10].

The robustness was evaluated using a 37-image corpus under multiple key sources. Using both single-source keys (Rand-projection-only or Quaternary-only) and a fused key (concatenating independent PUF outputs before hashing), all ciphertexts display near-ideal statistics-entropy ≈ 8

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bits per pixel, horizontal/vertical adjacency correlations ≈ 0 and strong diffusion under nonce changes (NPCR ≈ 99%, UACI ≈ 33%). Integrity checks fail essentially 100% of the time when 1% of cipher pixels are tampered with confirming effective authentication. Central tendencies are statistically indistinguishable across key sources, while the fused PUF modestly tightened dispersion, reflecting improved key-source robustness without compromising simplicity. The single-image exemplars and the multi-image, multi-key analysis establish that MoS_2 RO-PUF keys can be harvested once, expanded per session, and used to encrypt and authenticate images with security properties consistent with modern industry-grade practice for embedded/edge systems (per-session keying, nonce uniqueness, and MAC-based integrity). When the identical PUF key is reapplied in the decryption process, the original image is reconstructed without any observable degradation, indicating that the key generation is reproducible and stable[44]. Any deviation in the key, such as that caused by a different illumination state, produces unrecoverable noise, demonstrating the PUF's inherent sensitivity to environmental perturbations and rendering partial key recovery infeasible.[45].

A hardware-lightweight, fully reversible encryption-decryption pipeline driven by PUF-derived keys is also demonstrated in Figure 4c. A 16×16 binary dot-matrix (HNU/SLAB) is encrypted with a keystream obtained either directly from ternary/quaternary PUF symbols (XOR-3/4) or after SHA-based whitening (SHA-3/4). In all cases, the key seeds a keystream expanded to the plaintext length and XORed with the image; regenerating the same keystream from the key restores the image exactly, with no interpolation or quantization artifacts. The encrypted patterns exhibit near-ideal statistical masking with ~50% HD to the plaintext, and legible spatial structure is suppressed while exact recoverability is preserved. As XOR is linear over GF (2), implementation is trivial in digital logic and scales linearly with the array size; replacing "keys in memory" with "keys from physics" removes a high-value attack target. Under the final authentication stage, all 30 repeated outputs per device were identical, giving an observed FRR = 0 under the exact-match criterion. For the 64-device set, corresponding to 2016 inter-device pairs, the measured FAR was $1/2016=4.96\times10^{-4}$. Based on the 72 h RO drift (1.04–1.36 kHz), only 2

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out of 256 RO-pair margins fall within the worst-case flip window, giving a conservative BER_{UB} = 0.78%, which supports robust key regeneration under drift.

The approach combines three advantages essential for practical secure displays and sensor tags: (i) Device uniqueness and unclonability. PUFs supply per-die secrets without storage, reducing key-exposure risk; (ii) Cryptographic whitening. Optional SHA mapping "flattens" symbol bias and short periodicity, improving resistance to statistical inference on small arrays; (iii) System simplicity and robustness. The same lightweight XOR path delivers both highly randomized ciphertext for publication and exact decryption for verification; session keys can be generated as PUF-seeded, nonce-salted streams to avoid key reuse. These features enable a compact, fabric-friendly security primitive that is easy to integrate, scales to larger formats, and maintains a clear trade-off: maximized obfuscation of content with zero-loss restoration on demand. Notably, the hashed keys exhibit complete structural decorrelation from the raw responses, consistent with a cryptographically secure one-way function, ensuring elimination of statistical bias in the PUF output while preserving deterministic decryption capability[46].

Anti-attack Test and Performance Analysis

To evaluate the modeling capability of the proposed PUF, machine-learning-based prediction is performed on the pre-hash 32-bit quantized response, rather than on the SHA-256-derived key. For each of the 64 fabricated PUF instances, RO-frequency features are measured under four experimentally implemented optical operating conditions (dark, red, green, and blue illumination) at identical electrical bias. These experimentally measured RO features served as model input, and the corresponding pre-hash binary response is used as the prediction target. The resilience of the MoS_2 RO-PUF-derived keys against machine learning (ML) modeling attacks is quantitatively assessed in Figure 5a-d. The evaluation emulates modeling attacks in which an adversary measures RO frequencies and trains a predictor for the secret key. Using all four spectral features (No-light/R/G/B) as inputs and three representative learners (MLP, LIBSVM, LIBLINEAR), the

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hashed keys (32-bit SHA derived from raw multi-ary codes) remain unlearnable with mean test accuracy remaining at chance levels(MLP at 50.9 ± 1.1%, LIBSVM at 50.6 ± 0.8%, LIBLINEAR at 50.2 ± 0.6%) across training-set sizes[47]. The MLP epoch curves further show increasing train accuracy (~70%) while test accuracy saturates at 52.5 ± 1.3%, evidencing overfitting without generalization. Similar results are observed for ternary→SHA32 keys (MLP at 50.3 ± 2.2%, SVM at 49.5 ± 2.2%, LR at 50.9 ± 2.3%), and for per-channel attacks where each optical channel is modeled in isolation (all at ~50-52%). These results indicate that the hashing stage effectively breaks the learnable mapping from analog frequency vectors to the final secrets-consistent with a "controlled-PUF" design in which only cryptographically processed outputs are exposed.

A controlled baseline, conducted by removing hashing and attacking the raw multi-ary codes, demonstrates that learning becomes feasible: with a single channel and 48 training devices, average test accuracy reaches about 70-74% for quaternary symbols; with all channels and ternary symbols, MLP achieves 88.1±1.1% (SVM/LR 71-73%). The sharp contrast between "raw" and "hashed" targets establishes that RO frequency features contain exploitable structure, while a one-way randomness extractor (SHA) effectively reduces the mutual information with the key to near-zero under practical learning budgets.

Classical delay-based PUFs (e.g., Arbiter, feed-forward, and even k-XOR variants) typically succumb to logistic-regression/SVM/NN attacks given sufficient CRPs, frequently achieving high (> 90%) prediction accuracy; many countermeasures trade area or energy for obfuscation while retaining residual learnability. In contrast, the MoS_2 RO-PUF with optical challenges and mandatory hashing exhibits chance-level predictability under the same attack tools and CRP budgets, while also supporting a larger challenge space via RGB perturbations. In practice, the hashed-key mode meets the expected robustness for device-level authentication including unpredictability against standard ML modeling, stability to environmental drift (keys originate from post-processed, entropy-rich responses) and extensibility to session-unique keys via salted hashing/KDF without re-enrollment. For deployment, on-chip hashing (or KDF) and rate-limiting

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of raw RO data are recommended to preserve the controlled-PUF property and prevent leakage of learnable CRPs[44]. This combination of native hardware entropy, optical reconfigurability, and post-processing robustness yields a PUF system capable of generating high-entropy keys that are reproducible, resistant to environmental drift, and demonstrably secure against conventional ML-based modeling attacks[13].

To benchmark the performance of the proposed MoS_2 RO-PUF against state-of-the-art hardware security primitives, its key metrics were compared with representative implementations, including CMOS SRAM PUFs[29], graphene FET PUFs[23], carbon nanotube PUFs[5], and previously reported MoS_2 FET PUFs[28] (Figure 5e & Supplementary Table 7).The five axes quantify (i) Uniqueness (inter-chip HD), (ii) Uniformity (bit balance, HW), (iii) Multiple entropy sources & multiple keys, (iv) Reconfigurability, and (v) ML attack resilience. All values are normalized to the ideal (outer polygon). Our MoS_2 RO-PUF integrates electrical ring-oscillator statistics with RGB optical perturbations and enforces a controlled output path via on-chip hashing/KDF. This stack yields near-ideal bit statistics (Inter-HD ~50%, HW ~50%) and chance-level learnability under standard modeling: for hashed keys, test accuracy remains ~50% for MLP/LIBSVM/LIBLINEAR even as train accuracy approaches ~70%, and this behavior holds for both quaternary→SHA32 and ternary→SHA32 targets as well as per-channel attacks. By contrast, removing hashing reveals learnable structure (for example, with single channel raw quaternary data and 48 training devices, the model attains about 70 to 74% test accuracy), confirming that the cryptographic extractor collapses the mutual information between measurable frequencies and the final secret. In contrast to SRAM/graphene/CNT/MoS_2 FET PUFs, our design matches incumbents on uniqueness and uniformity, while expanding the envelope in multi-entropy/multi-key capability, reconfigurability, and ML-attack resilience.

These properties strengthen the PUF's performance envelope, including uniqueness, uniformity, and resistance to modeling. Our MoS_2 RO-PUF provides on-demand, multi-key provisioning (2/3/4-ary keys) and optical reconfiguration, enabling session-unique secrets without

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nonvolatile key storage, and supporting scalable IoT authentication, secure boot, and anti-counterfeiting. The implementation enforces mandatory on-chip hashing/KDF with salting and restricts access to raw RO responses, preserving the controlled-PUF behavior that underpins near-ideal attack resistance, while RGB challenge schedules enlarge the CRP space without appreciable hardware requirements. Overall, the proposed MoS_2 RO-PUF demonstrates nearly ideal uniqueness, perfect uniformity, multimodal optical and electrical reconfigurability, and strong resilience to ML attacks capabilities not previously achieved in PUF architectures.

Discussion

We demonstrate a compact, optically reconfigurable MoS_2-based RO-PUF architecture that merges materials-level photonic control with a cryptography-aware key harvesting pipeline. Wafer-scale monolayer MoS_2 enables the fabrication of uniform and stable RO arrays, while wavelength-selective illumination enables dynamic entropy injection without hardware modification. This dual-entropy mechanism allows multiple, statistically independent keys to be generated from a single device under different illumination conditions. By employing a robust key harvesting strategy, including q-ary quantization, SHA-256 hashing, and multi-source fusion, we construct high-entropy 32-bit keys with tight statistical bounds and reproducibility. These keys are further expanded into session-specific cryptographic keys via nonce-salted HKDF with domain separation, supporting rapid key renewal and large challenge-response spaces. Security benchmarks confirm that post-hash outputs exhibit no exploitable structure: linear, kernel, and neural classifiers perform at chance levels, even under large-scale modeling attempts. Application-level demonstrations, including authenticated image encryption and tamper detection, reveal ciphertexts with strong decorrelation and integrity guarantees. Our results suggest the MoS_2 RO-PUF as a promising platform for material-intrinsic, light-driven security primitives for secure key provisioning, device authentication, and tamper-evident computing in intelligent edge systems.

Methods

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Material synthesis. Wafer-scale monolayer MoS_2 epitaxially grown on c-face sapphire by oxygen-assisted CVD approaches described in our previous work. Characterization of the MoS_2 films employed AFM (Asylum Research Cypher S) together with Raman/photoluminescence measurements (LabRAM HR Evolution, Horiba Jobin Yvon).

Device Fabrications. First, Ti/Au/Ti (3/10/2 nm) electrodes were defined on Si substrates with 300 nm SiO_2 by ultraviolet (UV)-lithography and electron-beam evaporation (EBE). AR-5350 photoresist was spin-coated at 4000 rpm and soft-baked at 105 °C for 4 min. A 20 nm Al_2O_3 gate dielectric was then grown by atomic layer deposition (ALD) at 300 °C. The monolayer MoS_2 film was subsequently released from the sapphire growth wafer in deionized water and transferred onto the prepatterned substrate using a poly (methyl methacrylate) (PMMA) support layer. Channel patterns and vias were formed by reactive-ion etching in O_2 and Ar/CHF_3 plasmas, respectively, using a commercial system (PlasmaPro NGP80, Oxford Instruments). Finally, Au/Ti/Au (10/3/30 nm) was deposited as the contact layer.

Device Characterizations. Electrical/optoelectronic characterization employed a Lake Shore CRX-4K probe station equipped with an Agilent B1500A parameter analyzer and a Keysight DSOX1204A oscilloscope. Red, green, and blue illumination was supplied by a continuously adjustable laser (OYSL SC-PRO 7) and introduced into the chamber through a glass window.

CRP of RO PUF. Challenge definition. As now stated in Fig. 1 and in Methods, we define the input challenge as a tuple $C = (\lambda, P_{opt}, S, V_{sel})$ where λ is the illumination wavelength ($\in\{\lambda_1, \ldots, \lambda_{N\lambda}\}$, P_{opt} is the optical power (or power density) level ($P_{opt}\in\{P_1, \ldots, P_{Np}\}$), $s\in\{1, \ldots, N_s\}$ indexes the illuminated region on the RO array, and V_{sel} encodes which RO outputs (or RO pairs) are routed to the key-generation pipeline. For a given C, the array produces a vector of RO frequencies that is processed through random projection, quantization, and optional hashing to yield a pre-hash binary codeword r(C) of length L We take the PUF response to be r(C); thus, each CRP is defined as (C, r(C)). Combining this with the optical and spatial dimensions, the nominal maximum

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number of distinct CRPs addressable by the architecture is $N_{CRP,\,max} = N_\lambda \times N_P \times N_s \times N_{sel.}$ To avoid bias introduced by within-chip oversampling, we perform a chip-balanced analysis. Specifically, two RO instances (or RO pairs) are randomly selected from each chip to construct a dataset with equal statistical weight per chip.

Machine-learning-based modeling assessment. To assess whether the underlying PUF response can be learned from measured RO signatures, machine-learning models were trained to predict the pre-hash 32-bit quantized response from experimentally measured RO-frequency features. The SHA-256-derived output was not used as the attack target, since cryptographic hashing is intended to remove directly modulable correlations. The dataset comprised 64 fabricated PUF instances, each measured under four optical operating conditions (dark, red, green, and blue illumination) at identical electrical bias. Repeated random hold-out evaluation was adopted, and all data from the same physical device were kept within the same split to avoid information leakage. Prediction performance was quantified by the bit-wise accuracy averaged over all held-out response bits.

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Funding

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Author contributions

R.Y., and G.Z. co-supervised the research; T.L. conceived the idea, designed the experiments. Y.Z., L.H., and H.Y. performed the growth of monolayer MoS_2 wafers. T.L., Q.G., and X.Z. performed

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device fabrications and measurements. Y.Z., L.H., and H.D. performed AFM, PL and Raman characterizations. T.L., Q.G., H.C., Y.L., Y.W., H.Y., N.L., R.Y., and G.Z. analyzed data. T. L. and R.Y. wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Data availability

All the data generated and analyzed in this study are included in the paper and supplementary information. The Source Data underlying the figures of this study are available at https://doi.org/10.6084/m9.figshare.31920540. All raw data generated during the current study are available from the corresponding authors upon request.

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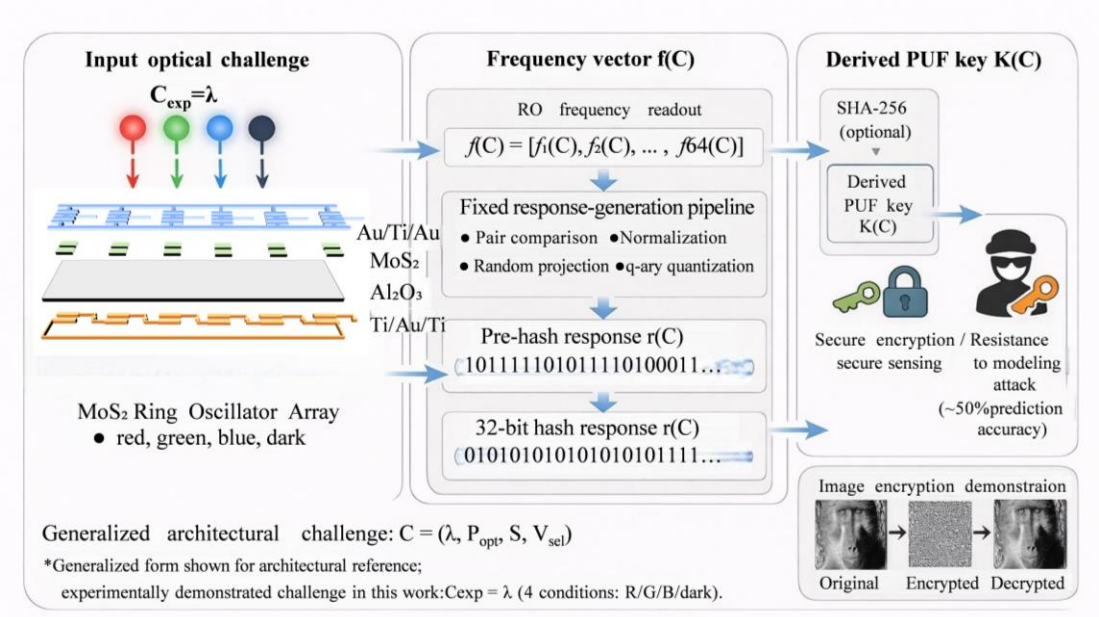


Figure 1. Schematic illustration of Secure Hash Algorithm (SHA) MoS_2 Ring Oscillator physical unclonable function (RO-PUF) cryptography. An applied challenge C modulates the MoS_2 Ring Oscillator (RO) array and generates a frequency vector f(C), which is converted by the response-generation pipeline into a 32-bit pre-hash response r(C). The challenge-response pair (CRP) is therefore defined as (C, r(C). In the experiments reported in this work, the externally varied challenge is the illumination wavelength (λ), whereas the optical power (P_{opt}), illuminated region (S), denotes selectable RO routing/RO-pair selection (V_{sel}), and readout configuration are fixed. In a generalized programmable implementation, the challenge can be extended to C = (λ, P_{opt}, S, V_{sel}). The optional SHA-256 stage produces the final derived key K(C). Baboon image in this panel is adapted from Rawpixel, licensed under CC0/Public Domain.

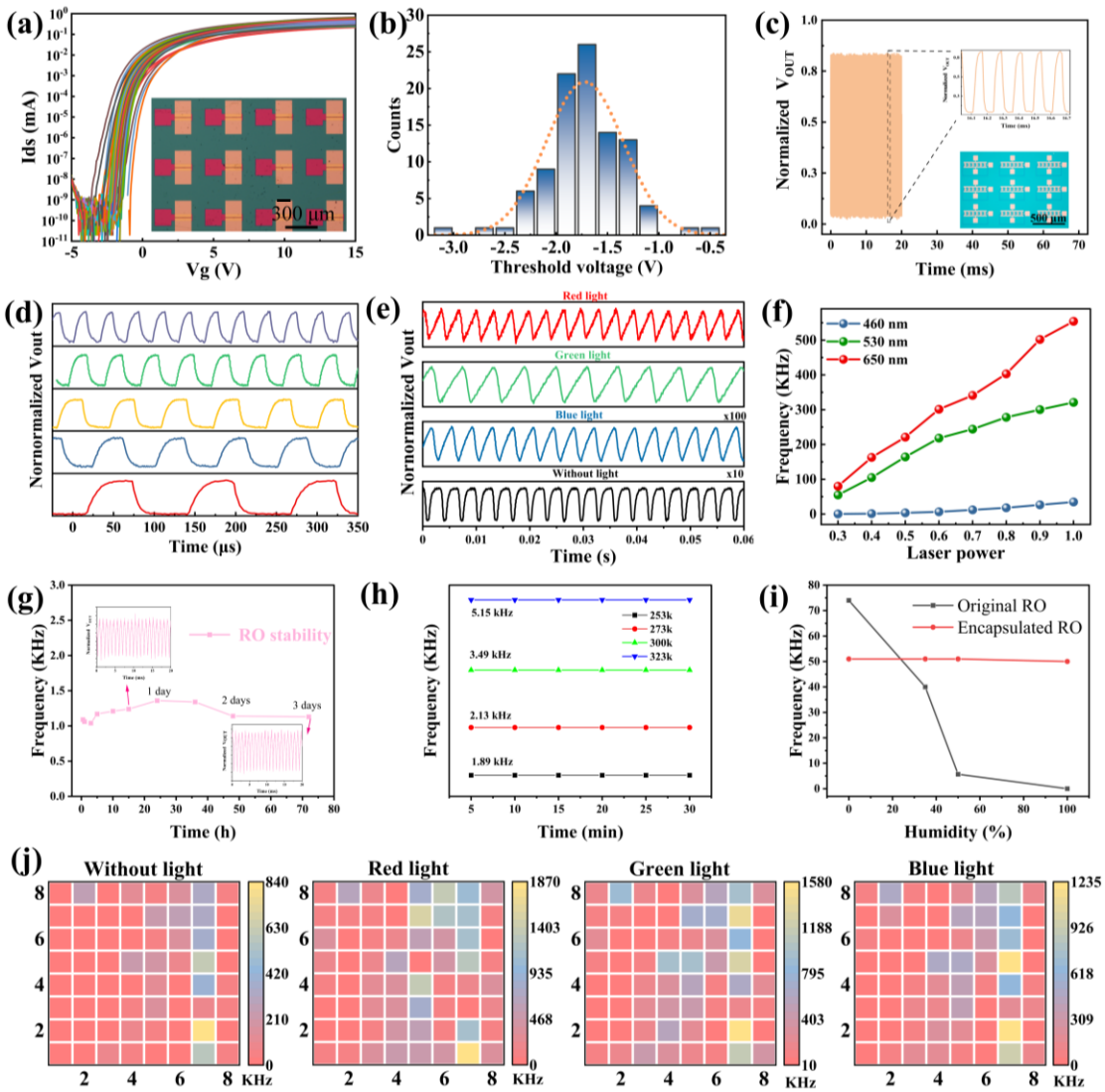


Figure 2. Performance characterization of the MoS_2 transistors and MoS_2 Ring oscillators. (**a**) Transfer curves of 100 randomly picked field-effect transistors (FETs) with the 20 nm Al_2O_3 dielectric. Bias voltage (V_{ds}) is 1 V, and the device yield is 100%. Here, Ids is the drain-to-source current and Vg is the gate voltage. Inset to (**a**) shows the enlarged image of the device array. Scale bar, 100 μm. Histogram and Lorentz distribution fits of the threshold voltage mean at -0.798 V, standard deviations at 0.0079 (**b**). (**c**) Single 5-stage RO performance. without light operated with 8 V. Inset to (**c**) shows the enlarged image of the MoS_2 Ring oscillators array. Here, Normalized V_{out} is normalized output voltage. Scale bar, 500 μm. (**d**). Output signals of a 5-stage RO operated without laser and with supply voltage from 1 to 9 V. (**e**) Output signals of a 5-stage RO operated without laser and with different laser (red, green and blue) at 30% laser power. (**f**) Output signals of a 5-stage RO operated with different laser (red, green and blue) and laser power from 30% to 100%. (**g**) Long time stability of a MoS_2 RO. Inset to (**g**) shows the are the test output waveforms on day 1 and day 3. (**h**) Temperature-dependent long-term stability of a MoS_2 RO. (**i**) Effect of humidity and encapsulation on RO frequency. (**j**) Frequency distribution of 64 5-stage MoS_2 Ring oscillators. without light, Red light (650 nm, 0.683 mW/mm^2), Green light

(530 nm, 0.174 mW/mm^2) and Blue light (460 nm, 3.39 μW/mm^2).

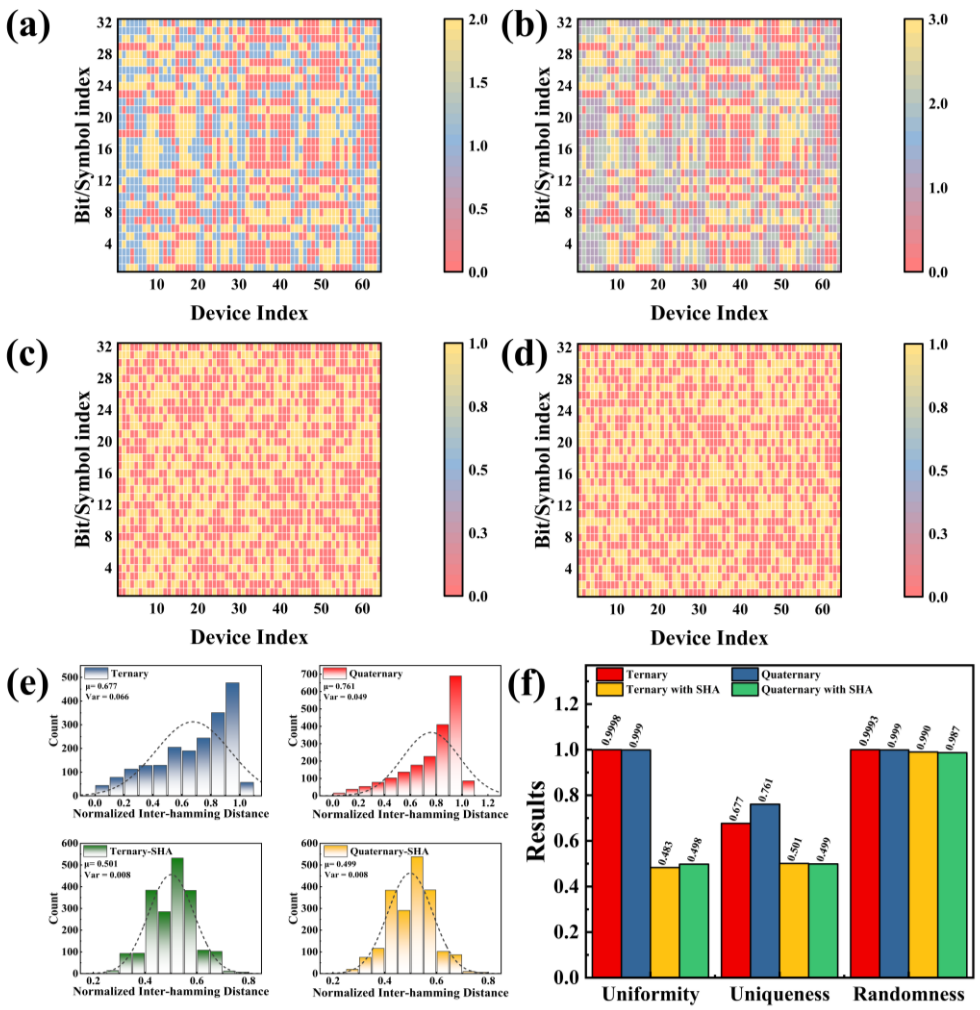


Figure 3. PUF pattern and performance comparison among of 64 5-stage MoS_2 Ring oscillators. (**a**) Ternary PUF key. (**b**) Quaternary PUF key. (**c**) Ternary-SHA PUF key. (**d**) Quaternary-SHA PUF key. (**e**) Statistical distribution of inter-Hamming distance for Ternary, Quaternary, Ternary-SHA, and Quaternary-SHA systems. The average values are (0.677, 0.761, 0.501, 0.499), and the variances are (0.066, 0.049, 0.008, 0.008). (**f**) Performance comparisons for the four PUF systems.

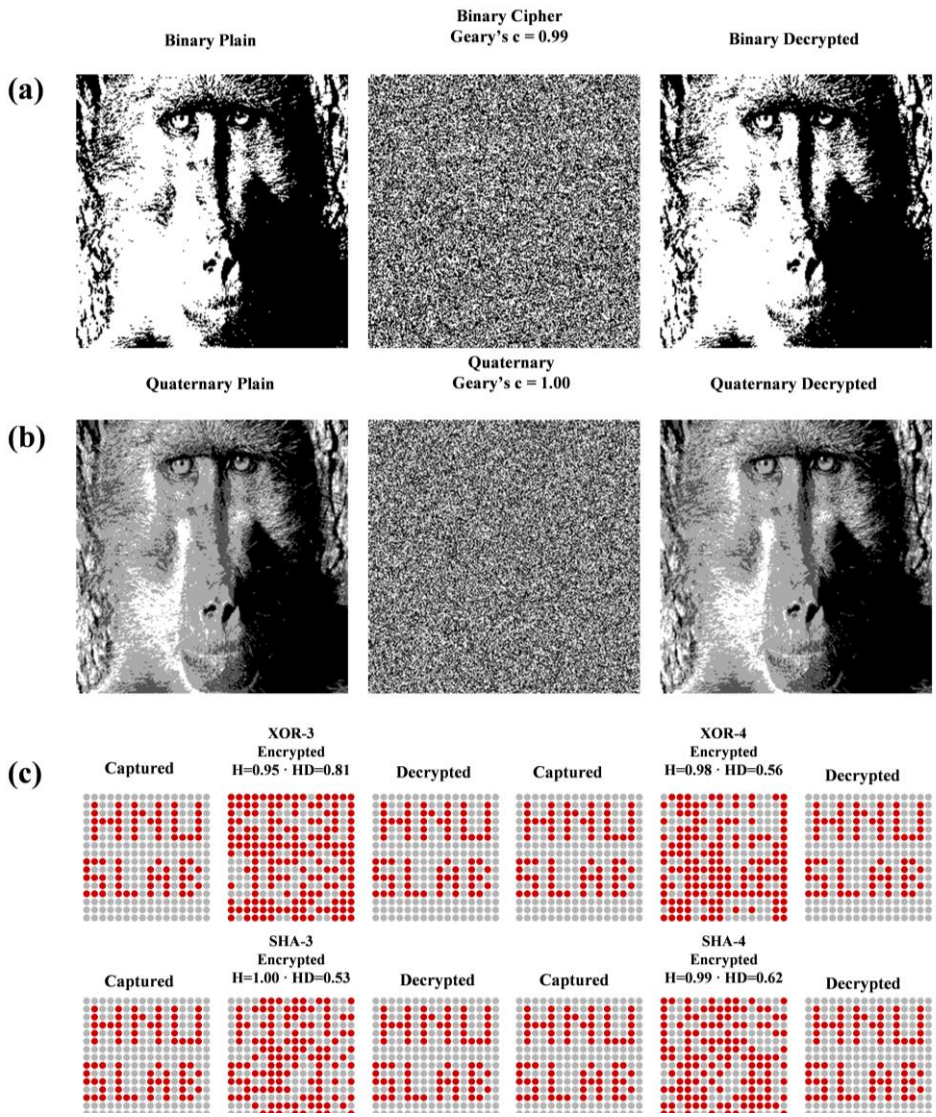


Figure 4. Demonstration and anti-attack test of PUF secret key encryption and decryption. (**a**) Graphic encryption and decryption with Binary PUF key. (**b**) Graphic encryption and decryption with quaternary PUF key. Baboon image in this panel is adapted from Rawpixel, licensed under CC0/Public Domain. (**c**) Demonstration of encryption and decryption of HNU and SLAB letter lattice.

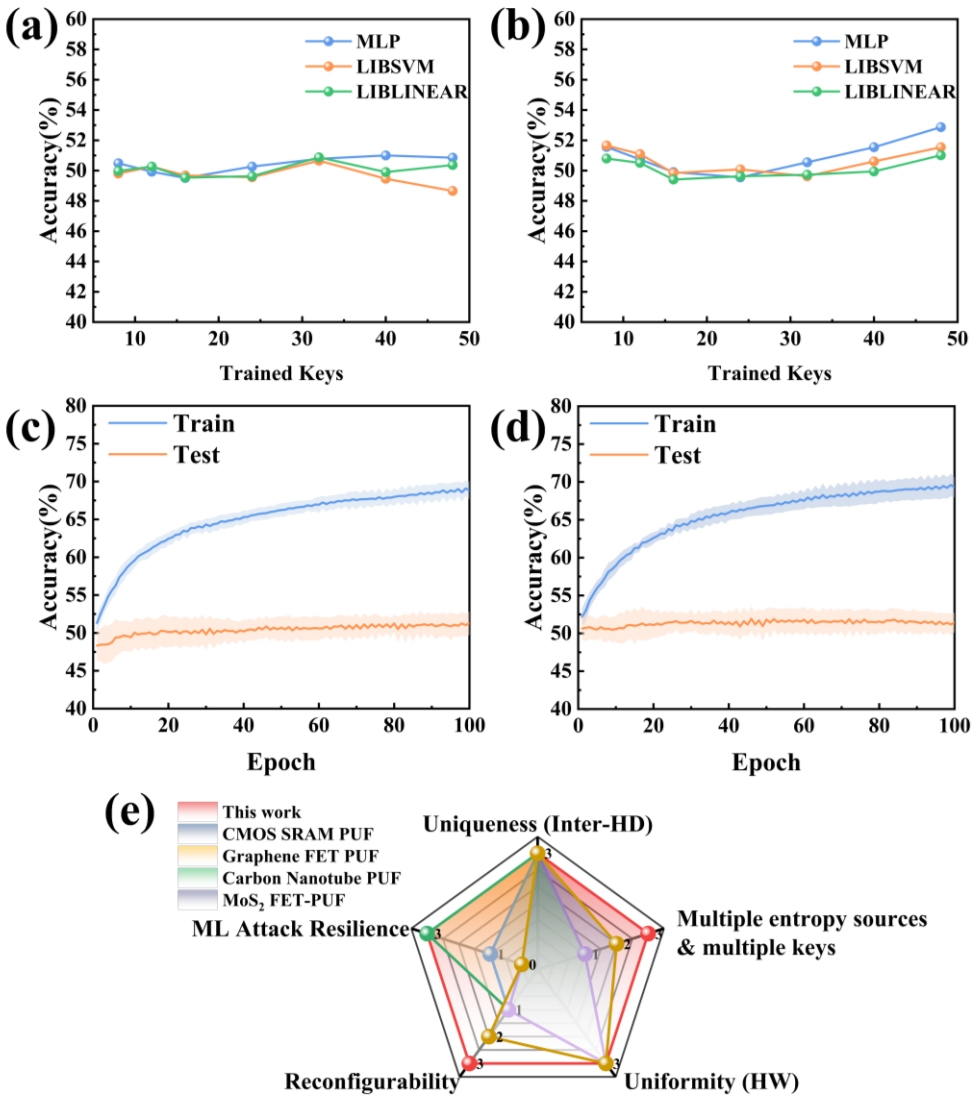


Figure 5. The anti-attack test and performance comparison of MoS_2 RO-PUF. (a, b) Bit-wise prediction accuracy of the pre-hash 32-bit response of the ternary and quaternary MoS_2 RO-PUFs, respectively, as a function of the number of training devices, evaluated using different machine-learning algorithms. (c, d) Prediction accuracy of the pre-hash 32-bit response of the ternary and quaternary MoS_2 RO-PUFs, respectively, as a function of training epoch using the MLP model. For (a-d), the dataset comprises 64 PUF instances measured under four optical operating conditions (dark, red, green, and blue illumination) at identical electrical bias. Train/test splitting was performed at the device level to avoid data leakage. Accuracies are reported as the bit-wise accuracy over the held-out test responses (mean ± s.d. across repeated random splits). (e) Performance comparison of the MoS_2 RO-PUF with previously reported PUF implementations[5, 23, 28, 29].

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Editorial Summary

Conventional Si-based physical unclonable functions (PUFs) are normally limited by limited entropy and lack of reconfigurability, exposing them to machine learning attacks. Here, the authors report optically reconfigurable PUFs based on 2D MoS_2 ring-oscillator arrays, demonstrating machine-learning-attack-resistant image encryption.

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